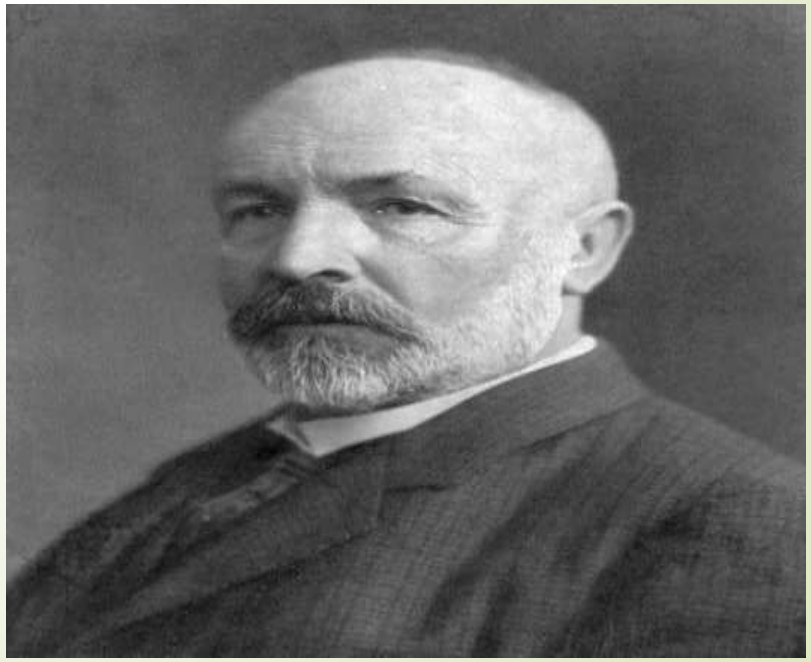
Sets

Sets

- is an unordered collection of object or a collection of elements
- **Set Theory**; created by a German Mathematician born in Russia named **Georg Ferdinand Ludwig Philip Cantor**
- Sample:

  {apple, mango, grape, orange}

  {2, 4, 6, 8, 10}



Sets

- objects in a set are also called the **elements** or **members** of the set.
- a set is said to contain its elements
- Describing a set:
  - {x, y, z}
  - P = {c, c++, java}
  - E = {2, 4, 6, 8, 10, 12, 14}
  - {1, a, Ted, Beth, Philippines}
  - {1, 2, 3, ...150}

Sets

- Two sets are equal if and only if they have the same elements
  - {a, b, c} = {c, a, b}
  - {1, 2, 3, 3, 4, 5, 5, 6} = {1, 2, 3, 4, 5, 6}
- Set Builder Notation
  - O = { x | x is an odd positive integer less than 15}
  - R = { x | x is a real number}
  - V = { x | x is a vowel }

Venn Diagram

- developed by **John Venn**
- method of using diagrams to illustrate set theory
- the Universal Set U contains all objects under consideration.



Venn Diagram

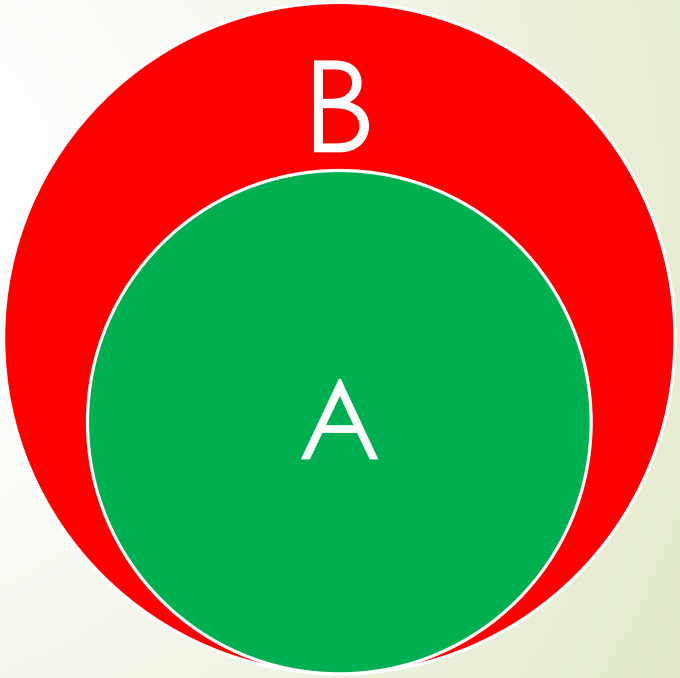
- represented using a rectangle.
- inside rectangle, circles or other geometric figures are used to represent sets.
- sometimes points are used to represent the particular elements of the set.
- often used to indicate the relationships between sets.
- *Draw a Venn diagram that represents V, the set of vowels in the English alphabet:*

Venn Diagram

- another notation to describe membership in sets by writing a $\in$ (element of) and $\notin$ (not an element of)
- Set with no elements is called the empty set or null set, denoted by $\varnothing$ or {}
- singleton set; {$\varnothing$}
- The set A is a subset of B if and only if every element of A is also an element of B.

Venn Diagram

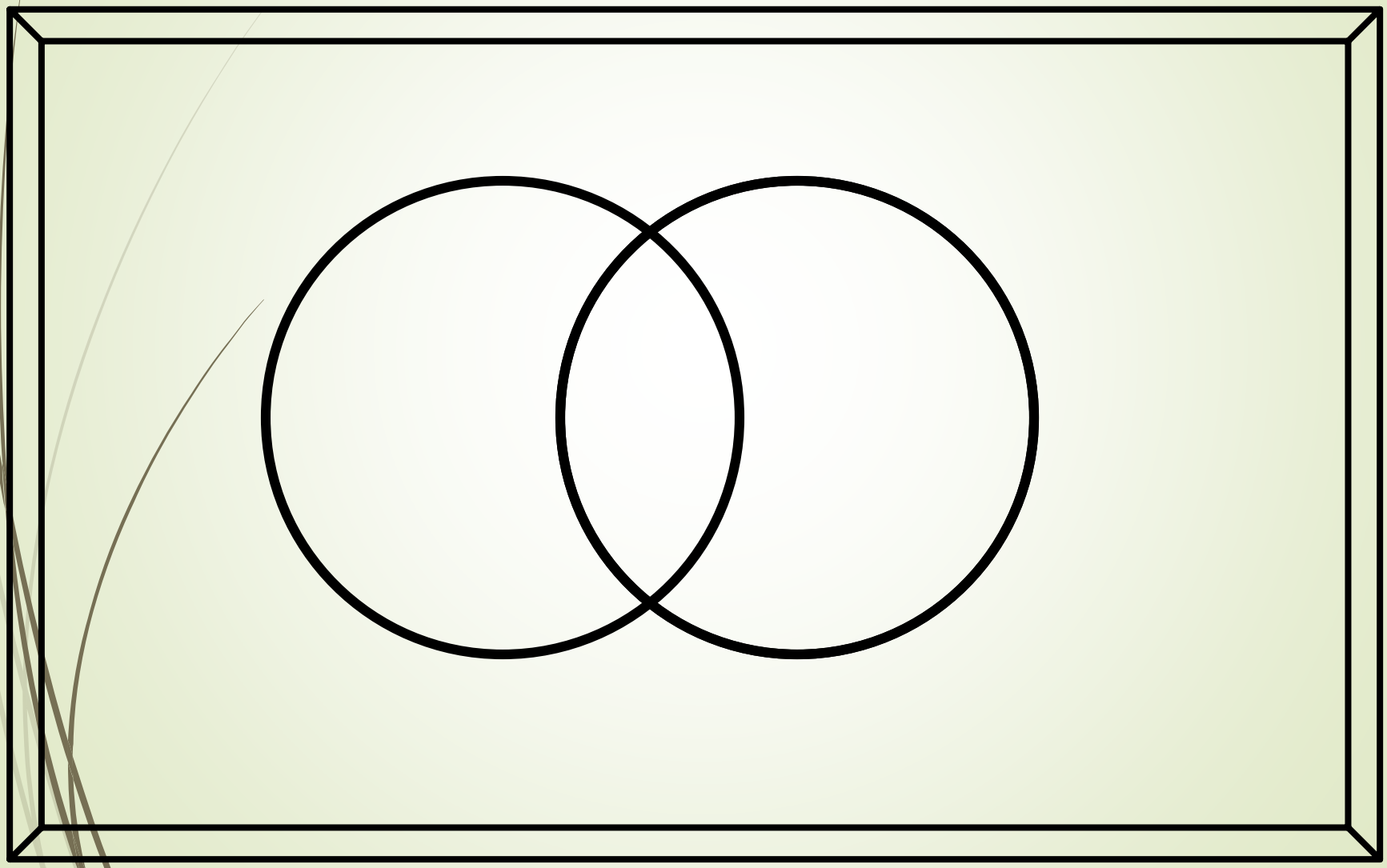
- $A \subseteq B$
  - A is a subset of B
- $B \supseteq A$
  - B is a superset of A
- $A \subset B$
  - A is a proper/strict subset of B
- $B \supset A$
  - B is a proper/strict superset of A



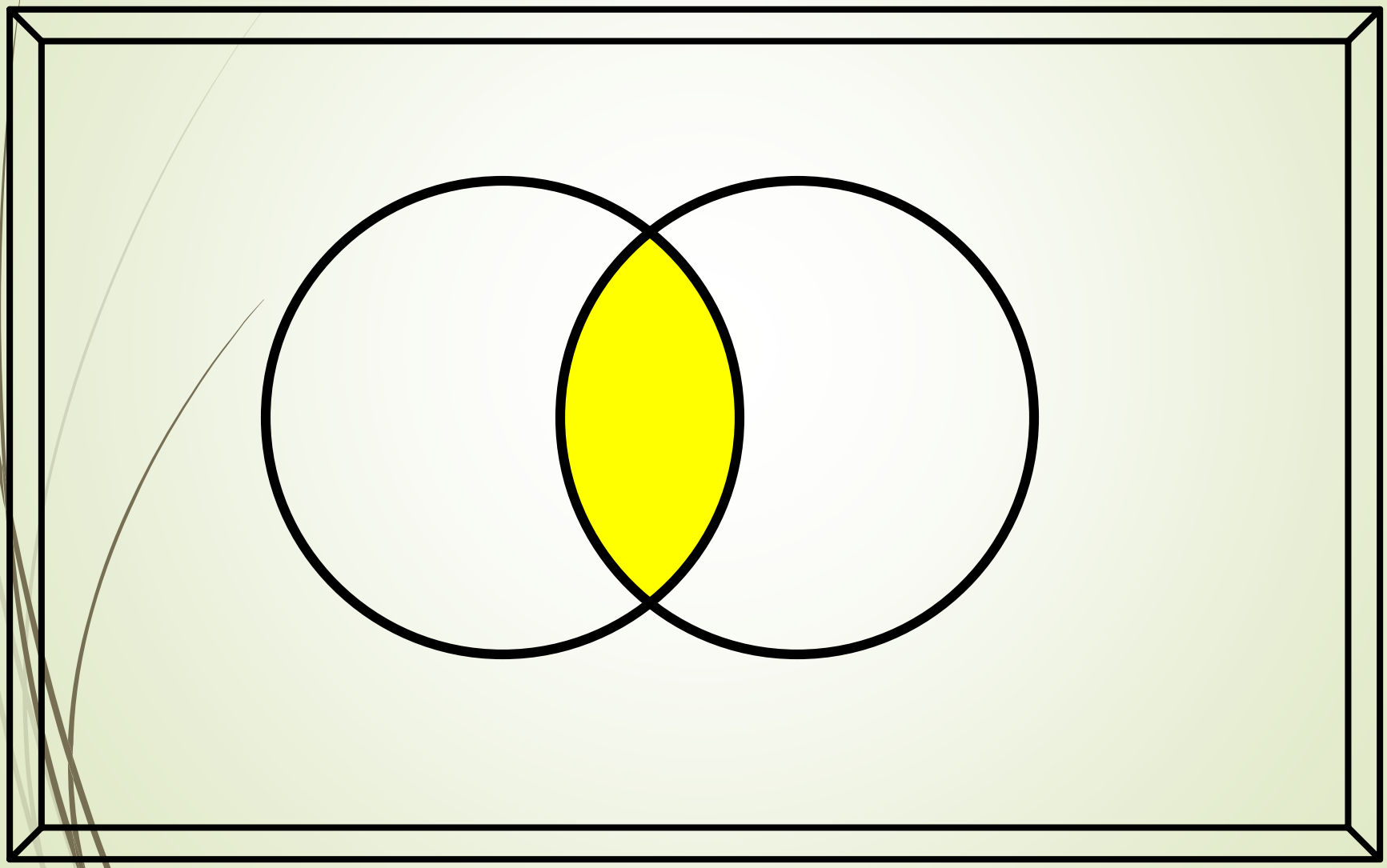
Venn Diagram

- Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a finite set and n is the **cardinality** of S.
- The **cardinality** of a set is a measure of the number of elements of the set. The cardinality of S is denoted by $|S|$. $|\varnothing|$
- A set is said to be infinite if it is not finite.
- Infinite sets may be countable or uncountable.

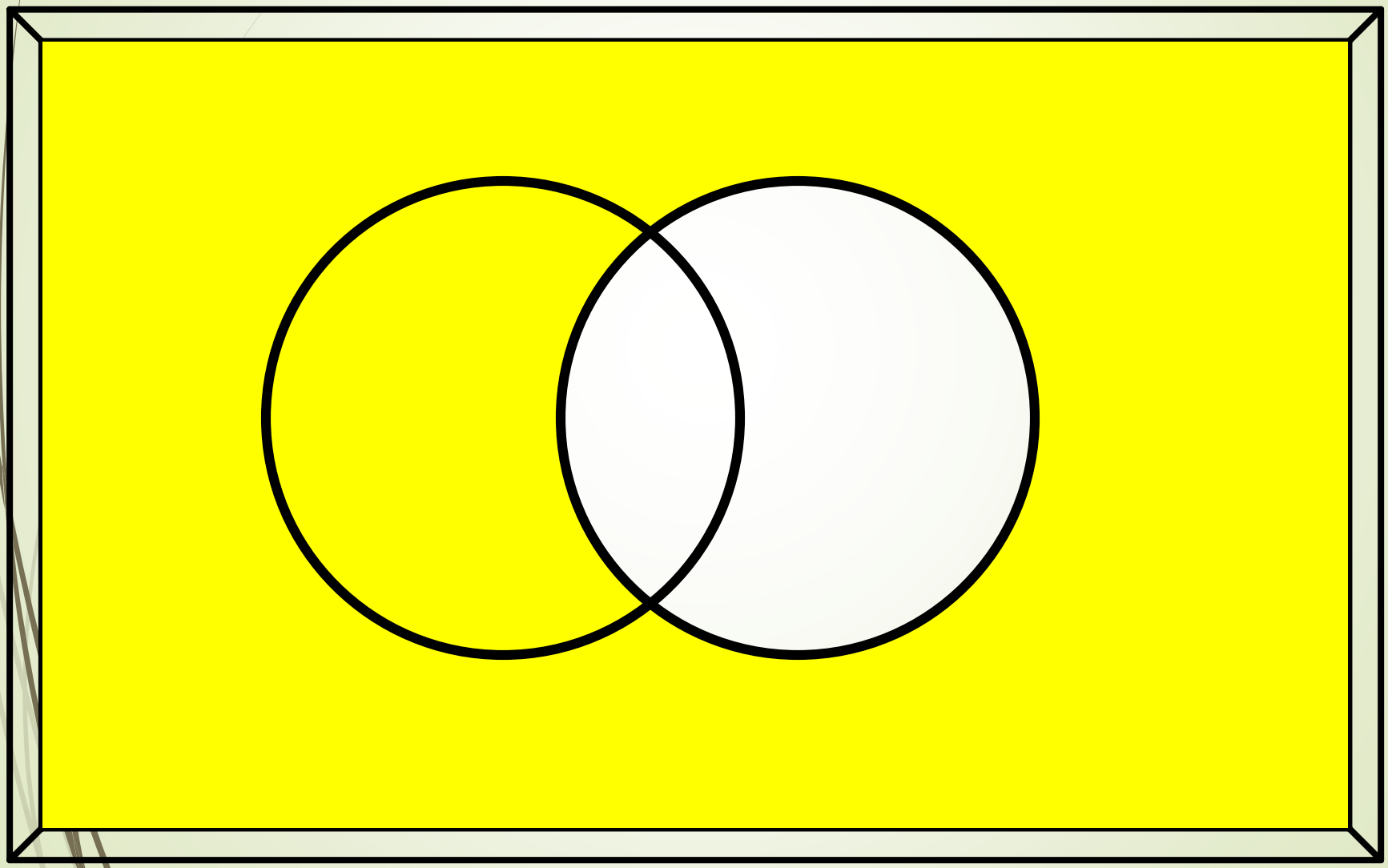
Union U



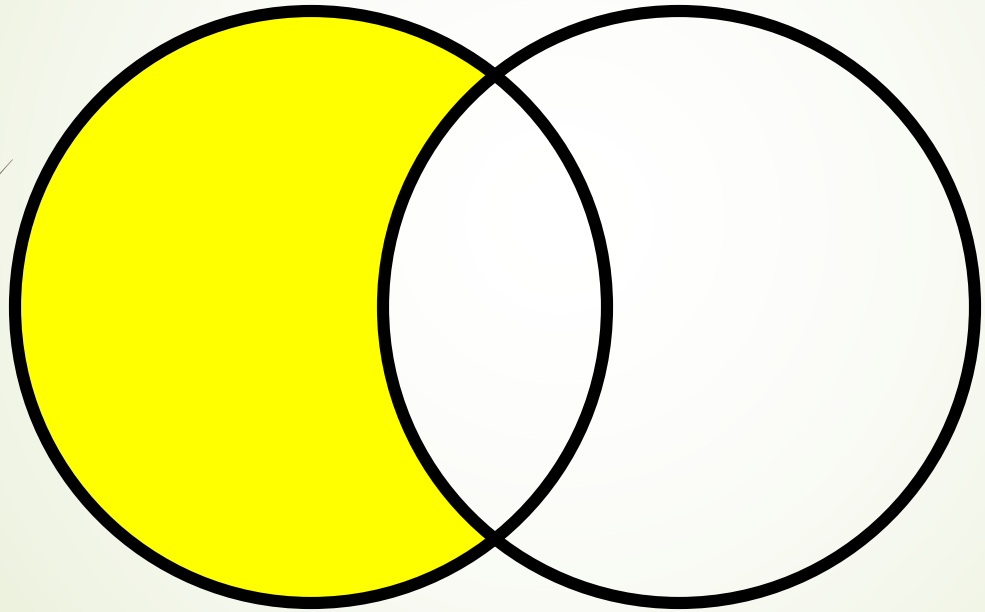
Intersection ∩



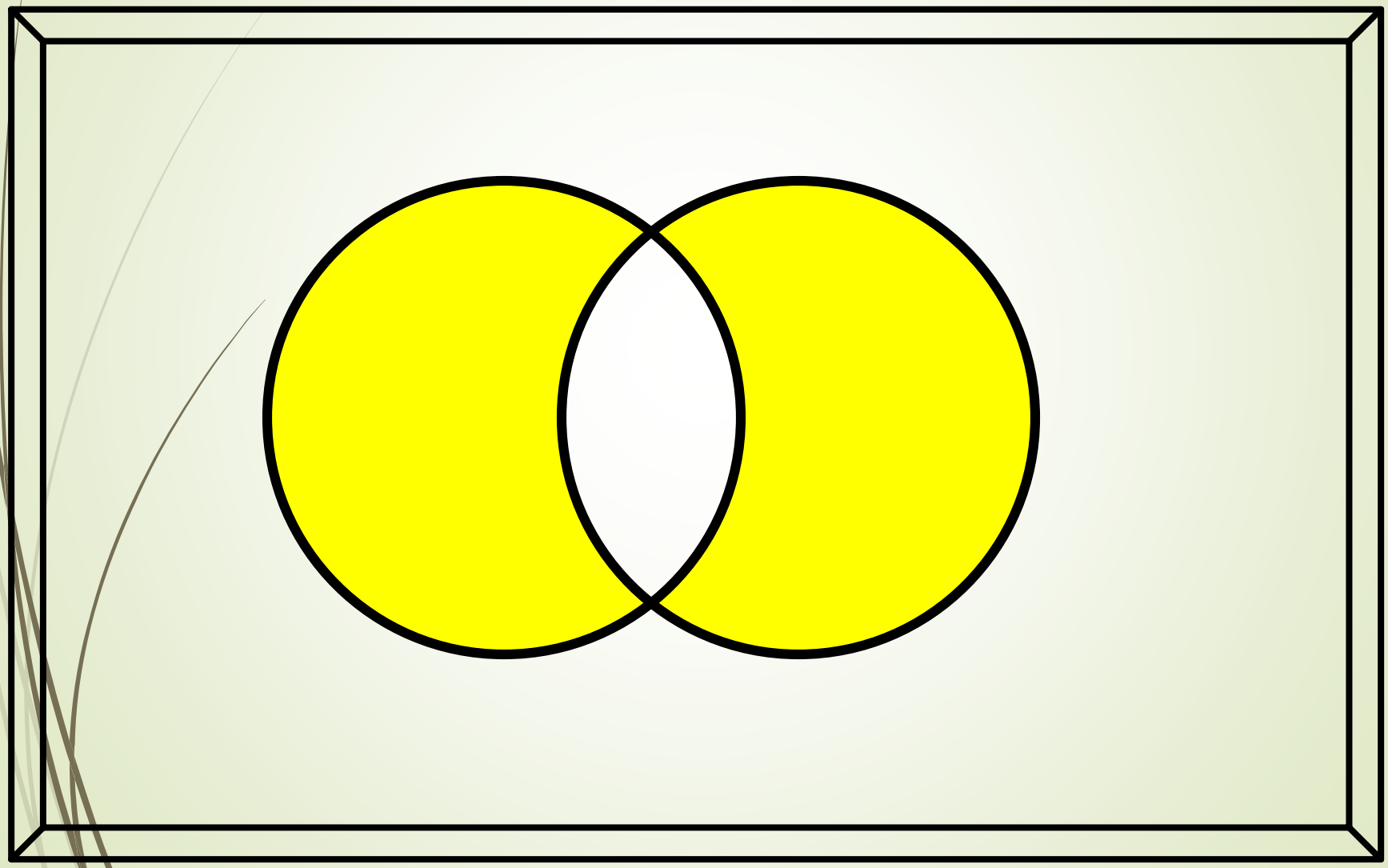
Complement ^{C}



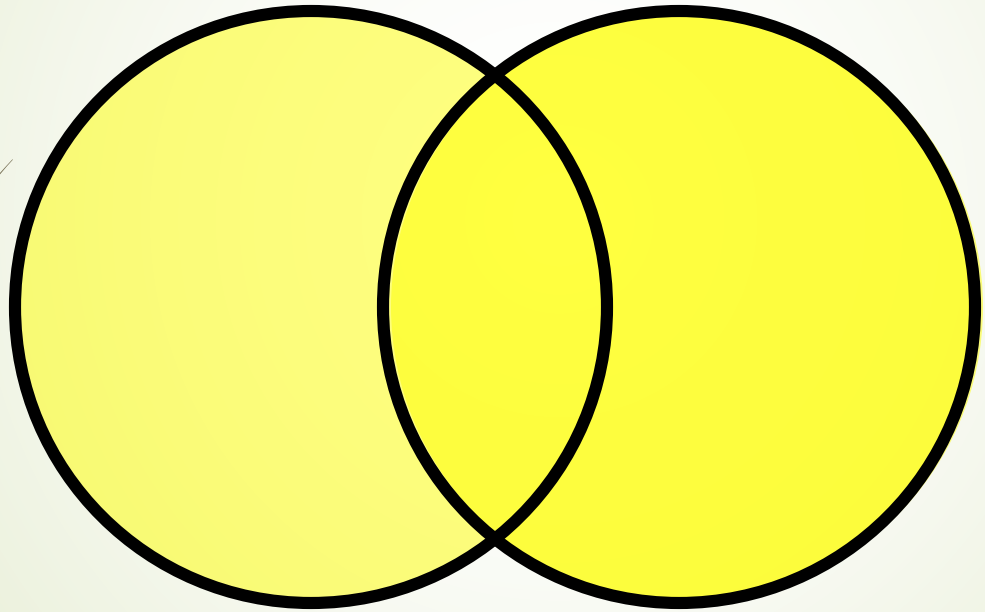
Difference -



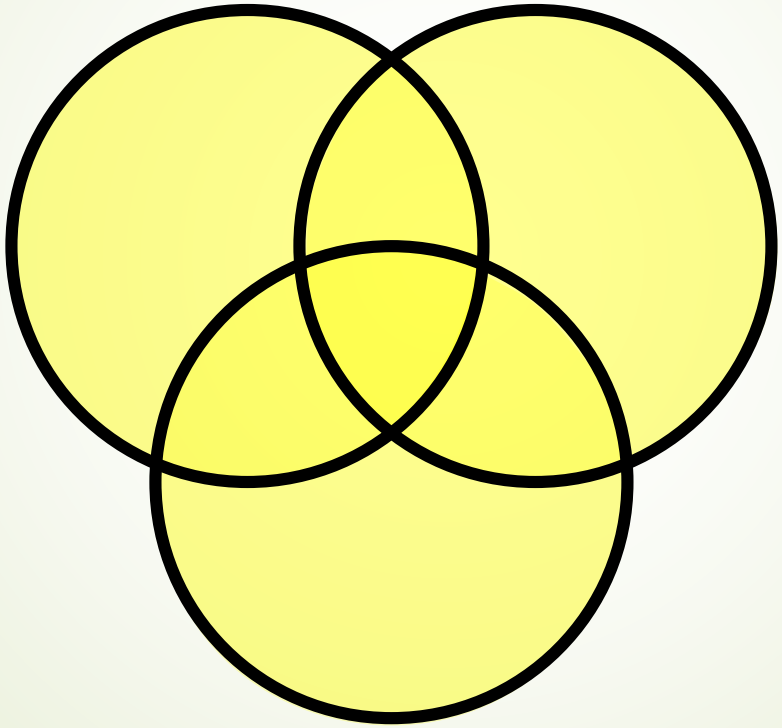
Symmetric Difference $\oplus$



Problems on Set



Problems on Set



Law on Sets

Laws	Equivalence
Commutative	$A \cup B = B \cup A$ $A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity	$A \cup \varnothing = A$ $A \cap \mathbb{U} = A$
Inverse	$A \cup A^C = \mathbb{U}$ $A \cap A^C = \varnothing$
Double Complement	$(A^C)^C = A$
Idempotent	$A \cup A = A$ $A \cap A = A$